

DALANZADGAD, Mongolia

WMO: 443730

Lat: 43.5772N

Lon: 104.4181E

Elev: 1465

StdP: 84.92

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.7	-22.7	-31.2	0.2	-20.4	-29.2	0.3	-19.5	14.0	-9.6	10.9	-7.5	2.1	290	0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	32.5	15.7	30.9	15.1	29.4	14.6	19.0	27.5	17.6	26.1	16.5	25.1	4.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	13.8	23.0	14.8	12.5	21.0	13.4	11.5	20.5	60.7	27.3	55.8	26.2	52.1	25.4	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	10.5	9.6	-26.9	35.2	2.4	1.8	-28.6	36.5	-30.0	37.6	-31.3	38.6	-33.1	39.9		
(5)			-27.2	19.6	2.4	2.3	-28.9	21.2	-30.3	22.6	-31.6	23.9	-33.3	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-13.6	-9.2	-1.1	8.1	14.5	20.2	23.2	21.2	14.9	6.1	-3.6	-11.2		
	DBStd	13.64	4.95	5.69	6.11	5.56	5.05	3.91	3.36	3.50	4.62	5.19	5.56	4.79		
	HDD10.0	2950	733	538	344	99	18	0	0	12	140	407	658			
	HDD18.3	4925	991	771	602	309	139	25	3	119	380	657	917			
	CDD10.0	1441	0	0	1	42	156	307	408	347	161	18	0	0		
	CDD18.3	373	0	0	0	1	19	82	153	102	17	0	0	0		
	CDH23.3	3556	0	0	0	11	194	801	1535	888	127	0	0	0		
	CDH26.7	1178	0	0	0	1	41	248	595	277	17	0	0	0		
(14) Wind	WSAvg	3.7	2.7	3.2	4.2	4.9	5.0	4.1	3.6	3.6	3.5	3.5	3.3	2.8		
(15) Precipitation	PrecAvg	121	1	2	4	4	11	20	30	30	12	4	2	2		
	PrecMax	221	7	11	16	17	54	78	100	76	50	22	9	17		
	PrecMin	67	0	0	0	0	0	1	1	0	0	0	0	0		
	PrecStd	34	1	2	4	4	11	17	22	20	12	5	3	3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	7.7	16.6	24.8	30.3	33.2	35.4	33.6	28.7	22.2	13.1	4.8		
		MCWB	-1.9	0.3	5.3	9.9	12.9	15.2	17.8	15.6	13.2	9.0	3.8	-1.6		
	2%	DB	-0.6	4.6	13.8	22.1	27.7	31.2	33.3	31.3	26.5	19.3	10.1	1.0		
		MCWB	-4.6	-1.5	3.7	8.4	11.7	14.7	16.5	15.2	12.2	7.7	2.1	-3.6		
	5%	DB	-3.6	2.4	11.2	19.8	25.6	29.5	31.6	29.5	24.6	16.9	7.7	-1.7		
		MCWB	-6.8	-2.7	2.4	7.4	10.8	14.0	15.8	14.6	11.4	6.6	1.0	-5.3		
	10%	DB	-5.9	0.0	8.6	17.5	23.4	27.7	30.0	28.0	22.8	14.7	5.2	-3.8		
		MCWB	-8.5	-4.2	1.0	6.4	9.8	13.2	15.2	14.2	10.7	5.6	-0.4	-6.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.7	0.9	6.3	11.4	16.7	19.4	21.5	20.1	17.0	10.3	4.4	-1.4		
		MCDB	2.2	6.9	14.4	21.7	26.2	29.1	30.5	26.5	25.3	18.6	12.3	4.2		
	2%	WB	-4.4	-1.2	4.2	9.4	13.6	17.1	19.6	18.2	14.8	8.5	2.6	-3.5		
		MCDB	-0.7	4.0	12.8	20.0	23.3	26.1	28.7	25.6	21.4	18.1	9.5	0.8		
	5%	WB	-6.6	-2.7	2.6	7.9	11.9	15.9	18.1	16.9	13.0	7.2	1.2	-5.2		
		MCDB	-3.6	2.1	10.7	18.5	22.6	24.7	27.0	24.9	21.3	16.1	7.5	-1.8		
	10%	WB	-8.4	-4.1	1.1	6.6	10.6	14.8	17.0	15.7	11.6	6.0	-0.2	-6.8		
		MCDB	-5.9	-0.1	8.4	17.1	22.2	24.3	26.0	24.2	20.8	13.9	5.1	-3.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	11.7	11.8	11.5	11.4	10.9	10.4	10.3	10.8	11.3	11.0	10.5		
		MCDBR	13.7	14.5	14.3	13.3	13.1	12.1	11.9	11.7	11.9	13.5	13.7	13.0		
		MCWBR	10.8	10.3	8.5	7.2	6.3	5.2	4.4	4.5	5.4	7.3	8.4	9.7		
		MCDBR	13.5	13.8	13.5	12.5	11.8	10.6	10.3	9.9	10.7	12.5	13.2	12.7		
	5% WB	MCWBR	10.6	10.1	8.4	7.4	6.8	5.8	5.3	5.0	6.0	7.3	8.3	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.289	0.333	0.377	0.398	0.413	0.397	0.382	0.345	0.312	0.275	0.255		
	taud		2.376	2.299	2.195	2.129	2.125	2.172	2.282	2.303	2.357	2.375	2.403	2.382		
	Ebn at Noon		888	906	900	885	873	859	869	871	880	868	855	854		
	Edh at Noon		76	99	126	147	152	146	129	122	107	91	74	69		
(44) All-Sky Solar Radiation	RadAvg		2.19	3.29	4.84	6.18	7.01	6.78	6.38	5.91	5.12	3.90	2.52	1.92		
	RadStd		0.22	0.34	0.28	0.28	0.35	0.37	0.31	0.26	0.22	0.13	0.21	0.22		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-2.22	N/A	-1.47	-1.82	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	-2.21	N/A	-1.34	-1.56	N/A	N/A	N/A	N/A

Nomenclature: See separate page